

PINAKI GHOSH

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Durham, NC | Work Authorization: F-1 CPT / OPT, Full-time Eligible

EDUCATION

Duke University | M.S. Interdisciplinary Data Science | GPA: 3.75/4.0 May 2027
Coursework: Causal Inference · Bayesian Statistical Modelling · Probabilistic ML Algorithms · Decision Modelling · Data Engineering · Deterministic Optimization

Indian Institute of Management Mumbai | PG Diploma in IT Management | GPA: 4.0/4.0

Indian Statistical Institute | Certification, Data Mining and Business Analytics

Institute of Engineering and Management, Kolkata | B.Tech, Electronics and Communication Engineering | GPA: 3.35/4.0

TECHNICAL SKILLS

Decision Science: Sequential Decision-Making · Causal Inference · Experimental Design · Pricing and Revenue Optimization · Human-Algorithm Interface Design

Predictive Modelling: Demand Forecasting · Segmentation · Propensity Modelling · Uplift and Incrementality · Recommendation Systems

Programming: Python (scikit-learn, SciPy, statsmodels, pandas) · R · SQL · Gurobi

Cloud and Tools: AWS (SageMaker, Glue, Athena) · Power BI · Matplotlib · Git

EXPERIENCE

ITC Infotech | Sr. Principal Consultant | Head, Customer Value Management Practice 2016–2020
Clients: Middle East Fashion Retail · South African Food and Fashion Retail · Nordic and Middle Eastern Airlines · Indian Luxury Hotel Chain (all ITC Infotech consulting engagements)
30-member team | \$4M annual practice revenue

- Designed the Customer Targeted Marketing Calendar. A treatment assignment engine mapping behavioral dimensions of 5 to 10 million loyalty customers against retailer objectives to generate individual-level promotion schedules and category-level pricing decisions.
- Delivered 1.5 to 2% experimentally verified incremental lift against holdout groups. \$20M+ in annual verified client revenue. \$4M practice payout contractually tied to verified lift.
- Developed trip reconstruction model for Nordic and Middle Eastern Airlines. Resolved multi-leg journeys across fragmented PNRs by trip rather than leg. Segmented by flying need and targeted journey leakage across carriers. Delivered 13% experimentally verified incremental lift.
- For Indian Luxury Hotel Chain: designed booking behavior segmentation by leisure, conference, and occasion-driven travel needs, booking timing, lead time, and frequency. Outputs informed pricing, packaging, and product design decisions. Connected behavioral need identification directly to revenue management strategy.
- Designed a customer analytics framework for affluent banking at a leading Middle East bank. Covered segmentation, product propensity, cross-sell, next best action, retention modelling, and CLTV estimation. Outputs delivered to relationship managers as decision intelligence for improving Average Daily Balance, a weighted daily measure of holdings. Observed 11% ADB improvement post-implementation.

ITC Infotech | Sr. Principal Consultant 2022–2024
Clients: Saudi Arabian Airlines Catering · UK Pharma

- Built demand forecasting pipeline for Saudi Arabian Airlines Catering. Random Forest model incorporating route, flight type, and seasonal features. Achieved more than 25% MAPE improvement and approximately 3% cost reduction. Adopted by catering managers as an operational decision tool.
- Designed UK Pharma personalization framework across prescription and OTC categories. Delivered 40% engagement uplift. Transitioned to client execution team for ongoing operation.

Marketboats Consulting | Head of Operations (Fractional Advisor)

2020–2025

- Designed B2B lead-generation pipeline, agentic in first principle. Covered ICP definition, profile matching, outreach sequencing, quality validation, and telemarketing decision support. LLM-based classification throughout with human override and monitoring layer.
- Delivered 70% throughput increase, 40% cost-per-lead reduction, and 60% reduction in manual intervention on strategic decision nodes.

ITC Infotech | Associate to Sr. Associate | Customer Value Management

2012–2016

- Built price elasticity, propensity, and behavioral segmentation models for health and wellness retail (NBTY). Delivered 40% improvement in campaign response and 25% increase in loyalty card activations.
- Designed trade promotion optimization model via causal attribution of promotional lift with long-term carryover effects. Delivered 30% ROI improvement for ITC Foods Biscuits through spend optimization rather than revenue uplift.

RESEARCH AND PROJECTS

Economic Value of Adaptivity (EVA) | Independent Research

2026–Present

- Designing SKUFlux: a four-layer RL-based adaptive retail allocation system grounded in EVA. Formally quantifies the gap between the theoretical maximum value of an optimal pricing policy and realized value when human decision makers act on algorithmic recommendations.
- The four layers span demand uncertainty quantification, causal demand driver decomposition, RL-based adaptive policy optimization, and representation design for human decision support.

When Does Demand Have Memory? | STA 611, Duke University

2026

- Empirically characterised AR(1) demand persistence across 45 Walmart SKUs (M5 dataset, 5.3 years). 91.1% show statistically significant persistence (mean $\rho = 0.550$). Confirmed ρ is a SKU-level property independent of product category (ANOVA: $F = 0.979$, $p = 0.384$).
- Chow tests reveal 78.3% of high-persistence SKUs exhibit structural regime shifts mapping to DE3 and DE4 demand environments in the EVA typology. Updated demand specification incorporates time-varying, store-adjusted persistence with heavier-tailed errors. Code: github.com/PinakiG-duke/sta611-demand-persistence

Cognitive-Centered Data Visualization | Bass Connections, Duke | Supervisor: Prof. Jana Schaich Borg

2025–26

- Two completed Prolific-based factorial experiments, $N=476$. Descriptive chart titles create systematic overconfidence in incorrect interpretations. Take-home titles dramatically increase correct conclusion rates.
- Findings challenge a foundational assumption in visualization guidance. Establishes empirical basis for representation design in adaptive decision interfaces. Paper in progress.

Multi-Objective Facility Location | Durham Public Schools, Duke Data+ | Graduate Research Lead

2026

- Extended capacitated facility location model across 851 planning units and 50 schools. Introduced Pareto frontier via epsilon-constraint method incorporating equity, stability, and distance trade-offs.
- Primary author. Presenting at Duke Data+ Symposium, July 2026.

Early Evidence on Generative AI Exposure and Labor Market Outcomes | IDS 702, Duke University

2026

- Difference-in-differences analysis using CPS ASEC data (2016-2025) linked to the AIIE industry-level AI exposure index (Felten et al. 2021). Post-2023 unemployment increased 0.29 percentage points in high-AIIE industries relative to low-AIIE, concentrated in workers aged 25-34.
- No comparable wage signal detected. Event-study coefficients confirm directional robustness. Findings inform early policy intervention targeting AI-exposed labour markets before displacement effects become conclusive.

Network Optimization | MOPTA 2026 Competition | Lehigh University | Individual Entry

2026

- Applying constrained optimization to a trucking network routing problem using Clarke-Wright savings heuristics and local search improvement. Demonstrates optimization formulation depth aligned to the SKUPlan layer of the EVA architecture.